A logarithmic-loss substitute for greedy attainment in abstract ring-variation spaces

Autonomous proof attempt for arXiv:2310.17309
`agent_lane_03` (GPT5.6)

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Status. Candidate partial result, likely valid; the source's uniform-equivalence question remains open. All implicit constants below are independent of the function and of the greedy rank.

Source question and setup

Gulgowski, Kamont, and Passenbrunner [1] study a p-normalized local orthonormal system $\Phi = (\phi_j)$ associated with a ν-ary filtration. For $1 < p < \infty$, it is a greedy (hence unconditional) basis of $X = L^p(S)$ and has the p-Temlyakov property. Given $0 < \sigma < p$, put

$$\alpha = \frac{1}{\sigma} - \frac{1}{p}, \qquad q = \min(\sigma, 2).$$

Their ring-variation space is equipped with

$$|f|_{V_{\sigma,p}} = \sup_{\Pi} \left(\sum_{R \in \Pi} E_p(f,R)^\sigma \right)^{1/\sigma}, \qquad \|f\|_{V_{\sigma,p}} = \|f\|_p + |f|_{V_{\sigma,p}},$$

where Π ranges over finite disjoint families of atoms or rings and $E_p(f,R)$ is the local best approximation error by S.

The open question is whether, under the paper's Bernstein/geometric hypothesis, one has uniformly in f and n

$$K(f, 2^{-n\alpha}; X, V_{\sigma,p}) \asymp \|f - \mathscr{G}_{2^n} f\|_p + 2^{-n\alpha} \|\mathscr{G}_{2^n} f\|_{V_{\sigma,p}}. \tag{1}$$

(The source reverses the subscripts once on the left of its display (4.3); $V_{\sigma,p}$ is plainly intended.) Here $K(f,t;X,V_{\sigma,p}) = \inf_{g \in V_{\sigma,p}} (\|f - g\|_p + t\|g\|_{V_{\sigma,p}})$.

Remark 4.3. *Recall the notation S_j for the space of functions that are, locally on atoms of $\mathscr{F}_j$, given by functions contained in S. Moreoever, P_j denotes the orthoprojector onto S_j. Corresponding to the local orthonormal basis Φ, it is natural to consider weak greedy algorithms corresponding to whole blocks $(P_j - P_{j-1})f$ instead of projections onto single functions $\langle f, \varphi_n\rangle\varphi_n$ for $\varphi_n \in \Phi$.*

To wit, let Γ_j be the set of indices n such that φ_n is contained in the range of $W_j = P_j - P_{j-1}$. Consider $\mu_j(f) = \max_{n\in\Gamma_j} |\langle f, \varphi_n\rangle|$ and choose a parameter $0 < t \leq 1$. Consider a weak greedy algorithm given by a sequence (j_k) such that

$$\min_{1\leq \ell\leq k} \mu_{j_\ell}(f) \geq t \max_{\ell\geq k+1} \mu_{j_\ell}(f)$$

and set

$$B_k f = \sum_{\ell=1}^{k} W_{j_\ell} f.$$

The results of Lemma 4.1 and Corollary 4.2 then remain valid if $\mathscr{G}_n f$ is replaced by $B_n f$. For a proof of this fact, we refer to Appendix D.

We don't know whether the result of Corollary 4.2 extends also to the variation spaces $V_{\sigma,p}$ instead of approximation spaces, i.e. we don't know if we have the uniform equivalence

$$K(f, 2^{-n\alpha}, L^p(S), V_{p,\sigma}) \simeq \|f - \mathscr{G}_{2^n} f\|_p + 2^{-n\alpha}\|\mathscr{G}_{2^n} f\|_{V_{\sigma,p}} \tag{4.3}$$

for all n. For this, it would suffice that greedy approximation is bounded in variation space, i.e. $|\mathscr{G}_n f|_{V_{\sigma,p}} \lesssim \|f\|_{V_{\sigma,p}}$, cf. Lemma 4.1 with the choices $\mathcal{X} = L^p(S)$, $\mathcal{Y} = V_{\sigma,p}$, and $\Psi = \Phi$. Indeed, we will see in (4.9) that (1) of Lemma 4.1 is true and Proposition 3.3 (ii) implies (2) of Lemma 4.1. Therefore, in order to show (4.3), it suffices to show (3) of Lemma 4.1.

On the other hand, we next prove some embeddings between the variation spaces $V_{\sigma,p}$ and the approximation spaces $\mathcal{A}^\alpha_q$ (cf. Sections 4.1 and 4.2) that allows us to conclude a weak version of Corollary 4.2 for variation spaces. This result is given in Theorem 4.6. In the embedding result of Section 4.2 (and therefore also in Theorem 4.6), it is crucial to assume condition w2* from (2.4) on the geometry of the atoms $\mathscr{A}$ and the space S. Finally, Proposition 4.8 provides an example showing that boundedness of greedy approximation in variation space is not true unless we assume a condition similar to w2*. This suggests that such a geometric condition is somewhat natural in the above embedding results.

4.1. **The embedding of $\mathcal{A}^\alpha_q$ into $V_{\sigma,p}$.** We have the following result:

Theorem 4.4. *Let $1 < p < \infty$, $0 < \sigma < p$ and set $\alpha = 1/\sigma - 1/p$. Letting $q = \min(\sigma, 2)$, the approximation space $\mathcal{A}^\alpha_q$ embeds continuously into the variation space $V_{\sigma,p}$.*

In the proof we need the concept of (Rademacher) type ρ (see for instance [12]). We recall that a Banach space $\mathcal{X}$ is of type $\rho, 1 \leq \rho \leq 2$, if there exists a constant T such that

$$\mathrm{Ave}_\varepsilon \Big\| \sum_i \varepsilon_i x_i \Big\| \leq T\Big(\sum_i \|x_i\|^\rho\Big)^{1/\rho}, \qquad x_i \in \mathcal{X}, \tag{4.4}$$

where the average is taken over all choices of signs $\varepsilon_i \in \{\pm 1\}$. It is known that L^p is of type $\min(p, 2)$ and clearly, inequality (4.4) for $\rho = r_1$ implies inequality (4.4) for all $\rho = r_2 < r_1$.

Exact source passage: arXiv:2310.17309v2, PDF page 19, display (4.3).

Proof Intuition

Fix the set of the 2^n largest coefficients of f and decompose $f = g + (f - g)$ for an arbitrary competitor in the K-functional. The error part is harmless: unconditionality controls its coordinate projection in L^p, and the variation Bernstein inequality pays exactly $2^{n\alpha}$, which is cancelled by the outside factor $2^{-n\alpha}$. The structured part g lies at the weak endpoint $\mathcal{A}^\alpha_\infty$. Truncating any $\mathcal{A}^\alpha_\infty$ coefficient sequence to 2^n entries puts it in the strong endpoint $\mathcal{A}^\alpha_q$, at the cost of only $(n+1)^{1/q}$. The embedding $\mathcal{A}^\alpha_q \hookrightarrow V_{\sigma,p}$ then completes the estimate. Thus the missing endpoint boundedness is localized to a logarithmic-in-cardinality loss.

Partial theorem

Assume $0 < \tau < p$, set $\beta = 1/\tau - 1/p$, and suppose the source's condition $\mathrm{BI}(\mathscr{A}, S, p, \tau)$ holds. Choose $0 < \alpha < \beta$ and define σ by $\alpha = 1/\sigma - 1/p$; put $q = \min(\sigma, 2)$.

Theorem 1 (Pointwise near-attainment)**.** *There is $C \geq 1$ such that for every $f \in L^p(S)$ and every integer $n \geq 0$, if*

$$A_n(f) := \|f - \mathscr{G}_{2^n} f\|_p + 2^{-n\alpha}\|\mathscr{G}_{2^n} f\|_{V_{\sigma,p}}, \qquad K_n(f) := K(f, 2^{-n\alpha}; L^p(S), V_{\sigma,p}),$$

then

$$K_n(f) \leq A_n(f) \leq C(n+1)^{1/q} K_n(f). \tag{2}$$

Equivalently, the loss is $O((1 + \log N)^{1/q})$ for an N-term greedy approximant.

Proof. Fix n, write $N = 2^n$, $t = N^{-\alpha}$, and let $P_\Lambda f = \mathscr{G}_N f$, where Λ is the greedy index set selected from f. We use four results established in the source:

$$\|u - \mathscr{G}_N u\|_p \lesssim N^{-\alpha}\|u\|_{V_{\sigma,p}} \qquad (u \in V_{\sigma,p}), \tag{J}$$

$$\|v\|_{V_{\sigma,p}} \lesssim N^{\alpha}\|v\|_p \qquad (v \in \Sigma^\Phi_N), \tag{B}$$

$$\mathcal{A}^\alpha_q \hookrightarrow V_{\sigma,p}, \qquad V_{\sigma,p} \hookrightarrow \mathcal{A}^\alpha_\infty. \tag{E}$$

For (B), the source states the estimate first for its local dictionary $\mathscr{C}$; since every element of Φ is a sum of at most ν elements of $\mathscr{C}$, the displayed form follows after changing the constant. Estimate (J) is (4.9), and (E) comprises Theorems 4.4 and 4.5(B) of [1].

We also need the following direct consequence of the coefficient characterization of approximation spaces (Fact 2.2 there). For every finite Λ with $|\Lambda| \leq 2^n$,

$$\|P_\Lambda u\|_{\mathcal{A}^\alpha_q} \lesssim (n+1)^{1/q}\|u\|_{\mathcal{A}^\alpha_\infty}. \tag{3}$$

Indeed, let (a_m) and (b_m) be the decreasing rearrangements of the Φ-coefficients of u and $P_\Lambda u$. Then $b_m \leq a_m$ and $b_m = 0$ for $m > 2^n$. The weak endpoint gives $2^{j(\alpha+1/p)} a_{2^j} \lesssim \|u\|_{\mathcal{A}^\alpha_\infty}$. The ℓ^q coefficient formula, summed only over $0 \leq j \leq n$, proves (3).

Take arbitrary $g \in V_{\sigma,p}$ and set $h = f - g$. Since Φ is greedy, best-approximation subadditivity and (J) give

$$\|f - P_\Lambda f\|_p \lesssim \sigma_N(f; \Phi) \leq \|h\|_p + \sigma_N(g; \Phi) \lesssim \|h\|_p + t\|g\|_{V_{\sigma,p}}. \tag{4}$$

Coordinate projections are uniformly bounded on $L^p(S)$ because Φ is unconditional. Hence (B) yields

$$t\|P_\Lambda h\|_{V_{\sigma,p}} \lesssim tN^\alpha \|P_\Lambda h\|_p \lesssim \|h\|_p. \tag{5}$$

For the structured part, (E) and (3) yield

$$t\|P_\Lambda g\|_{V_{\sigma,p}} \lesssim t\|P_\Lambda g\|_{\mathcal{A}^\alpha_q} \lesssim t(n+1)^{1/q}\|g\|_{\mathcal{A}^\alpha_\infty} \lesssim t(n+1)^{1/q}\|g\|_{V_{\sigma,p}}. \tag{6}$$

The (quasi-)triangle inequality in $V_{\sigma,p}$, followed by (4)–(6), gives

$$A_n(f) \lesssim (n+1)^{1/q}\big(\|f - g\|_p + t\|g\|_{V_{\sigma,p}}\big).$$

Taking the infimum over $g \in V_{\sigma,p}$ proves the upper estimate in (2). Conversely, $P_\Lambda f$ is a finite linear combination of local basis elements and therefore belongs to $V_{\sigma,p}$; using it as a K-functional competitor proves $K_n(f) \leq A_n(f)$. $\square$

Verification, scope, and novelty status

The exponent check is exact: $tN^{\alpha} = 1$, while the coefficient block $0 \le j \le n$ has $n+1$ entries and hence costs $(n+1)^{1/q}$. No linearity of the greedy map is used: P_{Λ} is fixed only after the greedy set of f is chosen, and the same linear coordinate projection is then applied to g and h. The proof works for quasi-Banach $V_{\sigma,p}$ because only a two-term quasi-triangle inequality is needed.

The theorem does *not* prove the uniform upper estimate in (1); removing the scale factor is exactly the endpoint cancellation problem. Eight upgrade attempts were made. In particular, an exact finite-tree mixed-integer search found no growing instability on balanced dyadic trees of depths four and five; this computation is exploratory and is not used in the proof. Searches through the run indexes and bounded web searches by exact arXiv id/title and the phrases "greedy approximation", "ring variation", and "K-functional" found no later explicit resolution as of 12 August 2026. This is a bounded audit, not a priority claim.

Human-review recommendation. Check chiefly the endpoint coefficient estimate (3) with the source's normalization, and the passage from the $\mathscr{C}$-Bernstein estimate to an N-term Φ projection. These are the only nonliteral combinations of source results.

References

[1] J. Gulgowski, A. Kamont, and M. Passenbrunner, *Properties of local orthonormal systems, Part III: Variation spaces*, arXiv:2310.17309v2 (2024). In particular Definition 3.1, Proposition 3.3, Fact 2.2, and Theorems 4.4–4.6.